

Research and Applications

Bidirectional imputation of spatial GPS trajectories with missingness using sparse online Gaussian Process

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ABSTRACT

Objective: We propose a bidirectional GPS imputation method that can recover real-world mobility trajectories even when a substantial proportion of the data are missing. The time complexity of our online method is linear in the sample size, and it provides accurate estimates on daily or hourly summary statistics such as time spent at home and distance traveled.

Materials and Methods: To preserve a smartphone's battery, GPS may be sampled only for a small portion of time, frequently <10%, which leads to a substantial missing data problem. We developed an algorithm that simulates an individual's trajectory based on observed GPS location traces using sparse online Gaussian Process to address the high computational complexity of the existing method. The method also retains the spherical geometry of the problem, and imputes the missing trajectory in a bidirectional fashion with multiple condition checks to improve accuracy.

Results: We demonstrated that (1) the imputed trajectories mimic the real-world trajectories, (2) the confidence intervals of summary statistics cover the ground truth in most cases, and (3) our algorithm is much faster than existing methods if we have more than 3 months of observations; (4) we also provide guidelines on optimal sampling strategies.

Conclusions: Our approach outperformed existing methods and was significantly faster. It can be used in settings in which data need to be analyzed and acted on continuously, for example, to detect behavioral anomalies that might affect treatment adherence, or to learn about colocations of individuals during an epidemic.

Key words: missing data, GPS imputation, Gaussian process, digital phenotyping, smartphone

INTRODUCTION

Recent research has demonstrated that smartphone-based metrics of physical activity, inferred from passive data such as GPS, accelerometer, and gyroscope sensors, are significantly associated with obesity,¹ diabetes,² various cardiovascular diseases,^{3,4} and mental health.⁵ Among various aspects of physical activity, mobility is an important factor that has been associated with intervention success and quality of life in clinical research. For example, in a cohort of patients with amyotrophic lateral sclerosis, self-administered smartphone scale scores correlate highly, and with low variability, with clinic-based scores, and thus could be used in clinical trials.⁶ An-

other study found that increased pain is significantly associated with reduced mobility quantified using daily summaries of GPS data in patients with diseases of the spine.⁷ In a mental health cohort, a method for behavioral anomaly detection was proposed based on mobility features summarized from GPS data that could signal the need for an intervention before an escalation of symptoms and relapse occurred.⁸

Smartphones have advantages over wearable devices in both clinical and public health research. Although the market for wearable devices like smartwatches and bands is thriving, ownership remains much lower than that of smartphones.^{9,10} Additionally,

most people stop using their wearable devices after 6 months,¹¹ and raw data are rarely available from these devices. Given the widespread adoption of smartphones and the ability to extract raw data and design transparent algorithms for processing those data, health-related smartphone applications hold great promise as tools to prevent disease, prolong life, and improve quality of life on a population level. For example, mobile phone GPS data were used to estimate compliance with lockdown measures during the COVID-19 pandemic.¹²

A major challenge in collecting smartphone GPS data is battery consumption. A standard GPS sensor sampling scheme, such as that implemented on the Beibe Research Platform¹³ for smartphone-based digital phenotyping, alternates between an on cycle and an off cycle; sensor data are obtained during the former and the sensor effectively “sleeps” during the latter. In addition to this missingness by design, additional missingness occurs for behavioral or technological reasons. These additional layers of missingness are referred to as missing not at random. We developed a new imputation method that is agnostic of the nature of missingness; it attempts to fill gaps in the data based on previously observed data regardless of why the data are missing. Note that for missing-not-at-random data, some of the summary statistics, such as time spent at home, derived from the imputed trajectories may still be biased; however, the remaining bias is almost certainly smaller than the bias resulting from the use of linear interpolation, which assumes travel at a constant speed over the missing interval.

The first principled method¹⁴ of handling intermittent missingness in GPS data at the individual level was recently introduced. The method consists of simulating individual-level mobility traces based on previously observed GPS mobility traces, without reliance on any external data, and then summarizing the resulting mobility traces using various daily summaries. There are some important drawbacks to this method: (1) it has a high computational cost, (2) it gives rise to biased summary statistics under certain settings, (3) it distorts distances for individuals who travel long distances, and (4) it lacks mechanisms to avoid extreme or unlikely imputed trajectories, such as a zigzag route toward a destination.

To address these limitations, we adapted the method¹⁴ using sparse online Gaussian Process¹⁵ (SOGP). SOGP reduces the computational cost of the method in 2 important ways. First, there are many settings in which the goal is to analyze data as they are collected, rather than to perform a single retroactive data analysis at the end of the study. An example of this is anomaly detection, in which the goal might be to use daily data to direct an individual to take a certain action. In this setting, the previous method requires processing data from the first day up to the current day for each new day in the study. In contrast, our new method updates an existing data summary based on the information content of the new day, resulting in substantial savings in computation time. Second, the sparsity property of SOGP ensures that the computational complexity of the method is linear in sample size during the imputation phase. To reduce bias, we use “bidirectional imputation” to simulate the missing GPS trajectories with more flexibility. The former GPS imputation method¹⁴ makes use of the rectangular method¹⁶ that summarizes a time series of location coordinates on a 2-dimensional plane with consecutive linear displacements. Here, we augmented that approach to a 3-dimensional sphere, removing the need for location projection and providing a more accurate calculation of distance. Finally, we added several condition checks to take special scenarios into account and generate more reasonable trajectories in [Supplementary Appendix A](#). Our method is generalizable and can be

applied to data collected by various applications and devices in different settings. A further investigation of the trade-off between the accuracy of the algorithm and the extent of missing data is conducted using simulation.

MATERIALS AND METHODS

Inferring mobility measures from GPS traces with missing data

The former method¹⁴ takes sequences of triplets consisting of latitude, longitude, and timestamp, and generates imputed trajectories and summary statistics as outputs. GPS latitude and longitude coordinates are first projected to a 2-dimensional plane, and those coordinates are converted to a mobility trace defined by a sequence of flights and pauses according to the rectangular method.¹⁶ Flights are defined as segments of linear movement and pauses are defined as periods of time in which a person does not move. Curved movement is approximated by multiple sequential flights. The mobility trace is modeled as a sequence of events, in which an event is either a flight or a pause.

The approach to dealing with missing data is to impute missing flight and pause events by resampling from all observed events. A person’s true Cartesian location after projection at time t is $L(t) = (L_x(t), L_y(t))$. Consider a period of missing data in a mobility trace that starts at time τ_0 and ends at time τ_1 . $L(t)$ is unobserved over the time interval $[\tau_0, \tau_1]$ with $L(\tau_0)$ and $L(\tau_1)$ both known. The approach aims to closely approximate $L(t)$ by borrowing information from the observed mobility trace outside this interval. It first determines if a flight occurs with probability $\Psi(L_x(t), L_y(t), t)$, in which $\Psi(\cdot)$ involves the similarities between the current triplet, $(L_x(t), L_y(t), t)$, and every observed event. Once the decision is made, we sample a displacement $\Delta = (\Delta^x, \Delta^y, \Delta^t)$ from the all flights or pauses with a multinomial distribution, in which the probability is also derived from the similarities between $(L_x(t), L_y(t), t)$ and every corresponding event. Let $N(t) = \max \{k : \tau_0 + \sum_{b=1}^k \Delta_b^t \leq t\}$ be the counting process for the number of simulated events elapsed by time t and let t_b be the time of the b_{th} event in the simulated trajectory. The simulated trajectory, used as an estimator for $L(t)$, bridged so it starts at $L(\tau_0)$ and ends at $L(\tau_1)$, is:

$$\hat{L}(t) = \frac{t_{N(t)+1} - t}{t_{N(t)+1} - t_{N(t)}} G(t_{N(t)}) + \frac{t - t_{N(t)}}{t_{N(t)+1} - t_{N(t)}} G(t_{N(t)+1}),$$

where

$$G(t) = \frac{\tau_1 - t}{\tau_1 - \tau_0} \left\{ L(\tau_0) + \sum_{b=1}^{N(t)} (\Delta_b^x, \Delta_b^y) \right\} + \frac{t - \tau_0}{\tau_1 - \tau_0} L(\tau_1).$$

This bridging ensures that the flights retain the property of being straight lines, whereas bridging $G(t)$ directly would lead to curvature.

Extension to spherical geometry

Given that the surface of the Earth is roughly a sphere, when the previous method¹⁴ projects the latitude and longitude to a 2-dimensional plane, it essentially forces a curved area to become flat. If the area is large, long distances become severely distorted. First, the projected area in the previous method is an isosceles trapezoid by design, so the method assumes that the location of the individual is limited to one hemisphere. Second, if a subject takes a long-distance flight, such as one from San Francisco Airport (37.62N, 122.39W)

to Helsinki Airport (60.32N, 24.95E), the displacements around the departure and destination locations are stretched or compressed by the projection. As an illustration, assuming that the 2 locations are within the projected area, the great-circle distances from the San Francisco Airport to the Golden Gate Bridge, to Downtown Palo Alto, and to the Bay Bridge are 24,388 m, 27,648 m, and 20,322 m, respectively, whereas using the projected distance, these same distances are 16,662 m, 6317 m, and 23,419 m, corresponding to an upward or downward bias of 12% to 78%.

Our proposed solution is to use latitudes and longitudes directly for the imputation, which only requires the redefinition of the rectangular method and the kernel function. In the rectangular model, given 2 observed positions L_s (starting point) and L_e (ending point) taken at times t and $t + \Delta t$, respectively, the straight line between L_s and L_e is defined to be a flight if and only if the following conditions are met. (1) The distance between any 2 consecutively observed positions between L_s and L_e is greater than r (ie, there are no pauses during a flight). (2) When drawing a straight line from L_s to L_e , the sampled positions between these 2 endpoints are at a distance less than w from the line. The distance between the line and a position is the length of a perpendicular line from that position to the line. (3) For the next sampled position L'_e after L_e , positions and the straight line between L_s and L'_e do not satisfy conditions 1 and 2. A schematic of the rectangular method is shown in Figure 1.

To adapt the rectangular method to the sphere, we retained the principles of the method. Instead of checking if the points between 2 ends fall within the rectangle of width $2w$, we checked if they fall within a “belt” (the combination of 2 symmetric spherical zones) as shown in Figure 1, in which the central axis is the great circle determined by the 2 ends. We defined 2 functions to achieve this: the shortest distance function between any 2 given locations, and the shortest distance function between a point and the great circle determined by any 2 points. Let $L(\alpha, \beta)$ denote an arbitrary latitude-longitude coordinate pair; $L_s = (\alpha_s, \beta_s)$, $L_e = (\alpha_e, \beta_e)$ are the starting and ending point; the radius of the Earth is denoted with E . The first function is the great circle distance between the 2 points:

$$D = E \cos^{-1}(\cos \alpha_s \cos \alpha_e \cos(\beta_s - \beta_e) + \sin \alpha_s \sin \alpha_e)$$

The second function is the length of the perpendicular line from that position to the great circle. We converted the latitudes

and longitudes to 3-dimensional Cartesian coordinates by $x = E \cos \alpha \cos \beta$, $y = E \cos \alpha \sin \beta$, and $z = E \sin \alpha$. Let $\gamma_s = (x_s, y_s, z_s)$, $\gamma_e = (x_e, y_e, z_e)$, and $\gamma = (x, y, z)$ denote the Cartesian coordinates of L_s , L_e , and L , respectively, and now the distance formula is

$$w = E \left| \frac{\pi}{2} - \cos^{-1} \left(\frac{\gamma_e \times \gamma_s}{\|\gamma_e\| \|\gamma_s\|} \cdot \gamma \right) \right|,$$

where $\times$ is the cross product and $\cdot$ is the inner product. In the kernel function, we replaced the Euclidean distance with the distance defined here.

SOGP for GPS data

A GP is defined as a collection of random variables in which any finite number of the variables have a joint Gaussian distribution.¹⁷ Such a process is fully specified by the mean function $m(x)$ and the covariance function $k(x, x')$. If x and x' are close, their covariance tends to 1; on the contrary, if x and x' are far from each other, their covariance tends to 0. GP is very flexible in prediction due to the infinite dimensional nature of the function space, and it can also incorporate expert knowledge if given an educated prior; however, it is computationally expensive in practice.

Given a set of input-output observations (x_i, y_i) , $i = 1, 2, \dots, n$, a SOGP¹⁵ approximates the posterior distribution by sequentially updating it with a representative subset of observations. Instead of computing the posterior distribution from scratch on every new input for a real-time prediction, the predictive quantities, such as the posterior mean and the posterior covariance of the process at arbitrary inputs, can be expressed as a combination of a finite set of parameters that depend on the training data only. The result of the Bayesian update using a GP prior with mean function $\langle f_x \rangle_0$ and kernel $K_0(x, x')$ is a process with mean and kernel functions given by

$$\langle f_x \rangle_{\text{post}} = \langle f_x \rangle_0 + \sum_{i=1}^n K_0(x, x_i) q_i$$

$$K_{\text{post}}(x, x') = K_0(x, x') + \sum_{i,j=1}^n K_0(x, x_i) R_{ij} K_0(x_j, x')$$

The parameters q_i and R_{ij} are computed only once during the training of the model and are fixed when making predictions.

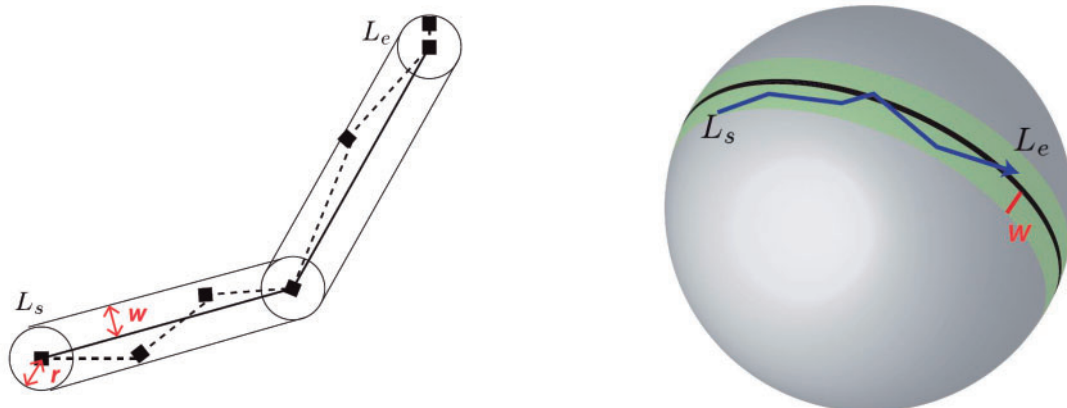


Figure 1. A schematic of the rectangular method (left) and the sphere method (right). In the rectangular method, the subject starts from L_s , makes a flight to the middle circle in 3 segments, then another flight to L_e , and stops in the last segment. All the segments that fall within the margins with width w are summarized as a single flight, and all the segments that fall within a circle with a radius of r are summarized as a stop. In the sphere method, the subject starts from L_s and makes a flight to L_e in 5 segments. All the segments that fall within the green “belt” with width $2w$ are summarized as a single flight. The central axis of the belt is the great circle determined by the 2 ends.

However, the online updating rule does not reduce the computational complexity. To further speed up the method, Csató and Oppel¹⁵ used only a subset of representative points (a.k.a. basis vector set or BV set) and established a rule that can add new representative points into the set and remove less representative points from the set using a sweeping process. Every new input is decomposed into 2 components: one component that is included in the subspace of the first t points and a second component that is orthogonal to the first. If the residual γ_{t+1} (orthogonal component) is greater than a threshold, the new input is added to the BV set, otherwise it is skipped. If the number of basis vectors exceeded the prespecified limit, the Kullback-Leibler divergence between the GP represented by all the basis vectors and every leave-one-out subset of these vectors is compared, and the one with the minimum score is removed. In either case, other parameters in the model are updated accordingly.

To apply SOGP to GPS data, we first converted a sequence of latitudes, longitudes, and timestamps to flights and pauses using the redefined rectangular method and calculated the corresponding mean of each piece as a triplet $z = (x, y, t)$. Among all the flight triplets, we then considered Y as a random variable and $\theta_T = (X, T)$ as features and assumed $Y = f(\theta_T) + \epsilon_T$ with $\epsilon_T \sim N(0, \sigma^2)$, where $f(\theta_T)$ is a GP with mean 0 and kernel matrix $K_0(\theta_T, \theta_T)$. Next, we applied SOGP with a prespecified size of BV and obtained m basis vectors. Finally, we switched Y with X to get another set of basis vectors. Note that the 2 sets may overlap and there are at most $2m$ basis vectors. We then repeated the same procedure on all the pause triplets, so that together there were at most $4m$ basis vectors. In our application, we set $m = 100$, but it can be varied as needed. For imputing missing trajectories, the empirical distribution function used for sampling can be constructed using those basis vectors only. Note that we did not use SOGP to predict the missing latitude and longitude based on the timestamp directly, but rather to select representative flights and pauses on the fly.

The kernel function $K_0(\cdot)$ incorporates both temporal and geographical information, and it is also periodic to reflect daily and weekly circadian rhythms with an amplitude that decays with time. The expression for $K_0(\cdot)$ is additive and consists of 3 terms:

$$K_d(z_1, z_2) = \exp\left\{-\frac{|t_1 - t_2|}{l_1}\right\} \exp\left\{-\frac{\left(\sin \frac{\pi|t_1 - t_2|}{c_1}\right)^2}{a_1}\right\},$$

$$K_w(z_1, z_2) = \exp\left\{-\frac{|t_1 - t_2|}{l_2}\right\} \exp\left\{-\frac{\left(\sin \frac{\pi|t_1 - t_2|}{c_2}\right)^2}{a_2}\right\},$$

$$K_g(z_1, z_2) = \exp\left\{-\frac{\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}}{g}\right\},$$

$$K_0(z_1, z_2) = \delta_d K_d + \delta_w K_w + \delta_g K_g, \quad \delta_d + \delta_w + \delta_g = 1.$$

Here, K_d is the daily component, K_w is the weekly component, K_g is the geographical component, and $l_1, l_2, c_1, c_2, a_1, a_2, g, \delta_d, \delta_w, \delta_g$ are hyperparameters. K_g reduces to $\exp(-d/g)$ if only either x or y is available, where d is the difference in the corresponding axis.

Bidirectional imputation

The previous method¹⁴ fills the missing trajectory by sampling a displacement Δ_i sequentially from the starting point L_s of the missing

interval to the ending point L_e . The last piece of the trajectory before the ending point must be a flight based on the bridging function and consecutive pauses are never imputed in the missing intervals. Note, however, that consecutive pauses may nevertheless occur: this is because an observed pause may precede a pause imputed in the beginning of a missing data interval or an observed pause may succeed a pause imputed in the end of a missing data interval. Given the fact that pauses take up more than 18 hours on average in a day,^{18,19} the ending point usually corresponds to a pause that may have started outside (before) the current on cycle. Hence, the algorithm may underestimate the duration of the pause at those locations. To address this issue, we lifted the constraint that no consecutive pauses are allowed during the sampling procedure and collapsed them into a single longer pause. Let $z_c = (x_c, y_c, z_c)$ be the current latitude, longitude, and timestamp; $z_i^p = (x_i, y_i, z_i)$, $i = 1, \dots, k_p$ be all the pause basis vectors; $z_j^f = (x_j, y_j, z_j)$, $j = 1, \dots, k_f$ be all the flight basis vectors; and $\sum \{\cdot\}^k$ be the operation of summing the top k largest numbers in the set. The probability that a flight happens at z_c is

$$p = \frac{\sum \{K_0(z_c, z_i^f)\}^k}{\sum \{K_0(z_c, z_i^f)\}^k + \sum \{K_0(z_c, z_i^p)\}^k}$$

To generate a trajectory, we first drew a random variable $B \sim \text{Bernoulli}(p^*)$ to determine if a flight or a pause happens, where p^* is a function of p , the remaining distance between the starting and ending points, and the remaining duration of the missing interval (the details are described in [Supplementary Appendix A](#)). Then, we sampled a displacement from all the representative flights/pause from a multinomial distribution with a probability $p_j = K_0(z_c, z_j) / \sum_{k_f/p} K_0(z_c, z_h)$. Moreover, instead of filling the missing trajectory solely in a forward direction, we alternated between proceeding forward and backward until 2 traces met each other so that the status before the ending point may be a pause. Let $L_s = (\alpha_s, \beta_s, \tau_s)$, $L_e = (\alpha_e, \beta_e, \tau_e)$ be the latitude, longitude, and timestamp of the starting point and ending point of a missing interval, respectively; let $L_f = (\alpha_f, \beta_f, \tau_f)$ be the current forward triplet; and let $L_b = (\alpha_b, \beta_b, \tau_b)$ be the current backward triplet. The algorithm starts from L_s , samples a flight or a pause, then locates the first stop L_f using the bridging function with L_s and L_e as the 2 ends. Next, the algorithm starts from L_e , also samples a flight or a pause, locates the last stop L_b but bridges with L_f and L_e as the 2 ends. The procedure continues forward and backward alternately, each time with a new starting or ending point, until L_f and L_b join, which is guaranteed by the design of the probability function p^* and the bridging function. After the completion of the imputation, consecutive flights with similar speeds and angles are merged, and consecutive pauses are merged. A schematic is shown in [Figure 2](#). The details are described in [Supplementary Appendix B](#).

RESULTS

Computational performance

We applied the previous method¹⁴ and our new method to a dataset collected as part of an ongoing study at the Massachusetts General Hospital. The study²⁰ sought to determine if smartphone GPS data could reveal differences in recovery after breast-conserving surgery and mastectomy. The size of observed data in 1 week was approximately 5040 triplets (timestamp, latitude, longitude), and the missing data duration was 154 hours on average across all individuals in the cohort. We replicated the procedure 5 times and calculated the average running time for each sample and as shown in [Figure 3](#), the com-

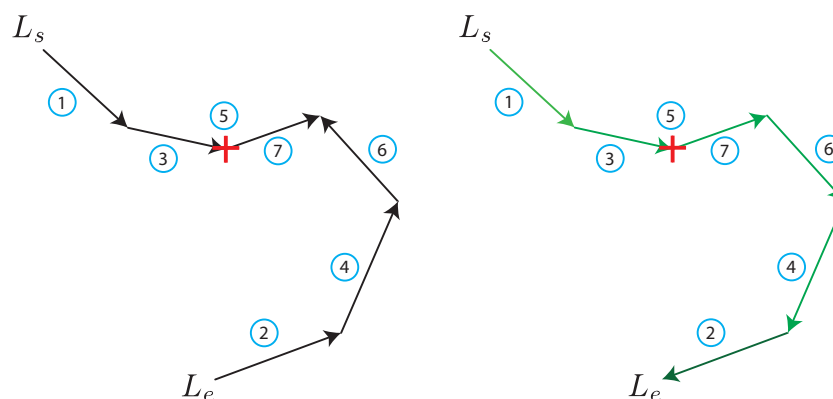


Figure 2. A schematic of applying bidirectional imputation. The missing trajectory starts from L_s and ends at L_e . The left panel shows the order of imputation: (1) forward flight—(2) backward flight—(3) forward flight—(4) backward flight—(5) forward pause—(6) backward flight—(7) forward flight. Then the 2 ends join and the algorithm stops. The right panel shows the final trajectory with adjusted directions and temporal orders. The arrows represent flights and the red cross represents a pause.

putational time was linear in sample size for our new method and approximately quadratic for the previous method. The new algorithm outperformed the previous algorithm after 12 weeks of observations.

Evaluation of basis vectors selected by SOGP

One of the key questions about applying SOGP is whether the selected basis vectors are representative of all observed flights and pauses. We used the same data here as in the previous section but chose not to use simulated trajectories because when the underlying data-generating mechanism is known, as is the case in a simulation, knowledge of that mechanism may give an unfair advantage in selecting good basis vectors. The left panel in Figure 4 describes a subject's observed 4185 flights and pauses during 5 months and the right panel shows all 320 selected flights and pauses, which almost cover every part of the observed trajectory. It demonstrates that the kernel function in our SOGP-based method is well designed and the parameters are properly chosen.

Quality of imputed trajectory

In this section, we examined a fully observed real-world GPS datasets. We randomly removed $(T - m)$ consecutive minutes of data every T minutes, where $T \sim N(\mu, \sigma)$, and where m , μ and σ are study-specific parameters, and kept the remaining as the observed data to mimic the sampling scheme of our Beiwu Research Plat-

form.¹³ Then we implemented 10-fold multiple imputation using our new method on the artificial observed data. To clearly present the imputed trajectories, only 1 of the 10 realizations is shown in the graphs. The data were collected using a dedicated GPS by G.L. for 3 consecutive days; the device was active all the time and it collected data at 1 Hz. As shown in Figure 5, our approach recovered the majority of the missing trajectories. This result is supported by Table 1, in which the means and standard deviations of the downstream summary statistics derived from the imputed trajectories over 10 replications are presented and compared with the truth. The 95% confidence intervals of both home time and flight time covered the truth. The distance traveled was often underestimated because of the simplifications of the trajectories. Another similar simulation with a synthetic dataset can be found in Supplementary Appendix C.

Accuracy of GPS-based colocation assessment for COVID-19 with and without imputation

It is natural to ask what the optimal sampling strategy is to obtain a reliable imputed trajectory while consuming as little battery power as possible. We consider this problem in the context of the ongoing coronavirus disease 2019 (COVID-19) pandemic. Based on the Public Health Guidance for Community-Related Exposure²¹ by the U.S. Centers for Disease Control and Prevention (CDC), individuals who have had close contact (<6 feet) with persons with COVID-19 for over 15 minutes (consecutively) should self-quarantine for 14 days. Assume that we have GPS data collected for 2 individuals at 1 Hz with a 15-minute on-off cycle and the proportion of the off-cycle to the entire cycle is p . Assume that one of the individuals tests positive for COVID-19 and has previously had multiple colocations with the other, defined as the 2 individuals being within 10 m from each other for at least 15 minutes. (This definition is more conservative than the CDC definition.). Our goal is to use simulation to explore how many of those colocations can be detected using the imputed trajectories from the proposed method under different rates of missing data. Distinct mobility patterns are simulated for the 2 individuals so that they are colocated at the same regularly visited place 6 times over 14 days and at an occasionally visited place once for h minutes over 14 days. The full trajectories are generated by first drawing a random starting time from the time range of the activity, then drawing a random speed based on the activity, and finally adding normally distributed noise to mimic measure-

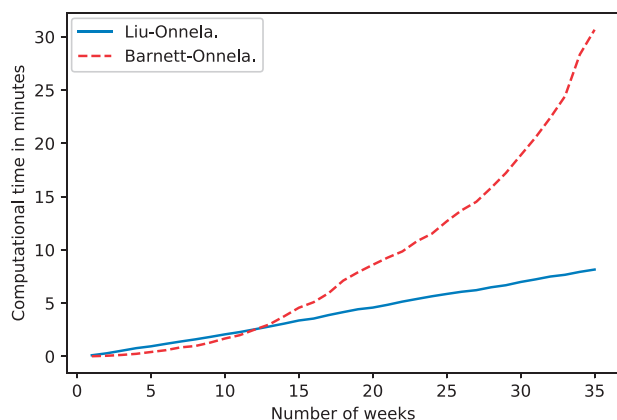


Figure 3. Average cumulative computational time for the new imputation method (Liu-Onnela) vs our previous imputation method (Barnett-Onnela) as the input sample size increases in 1-week blocks over 5 replications.

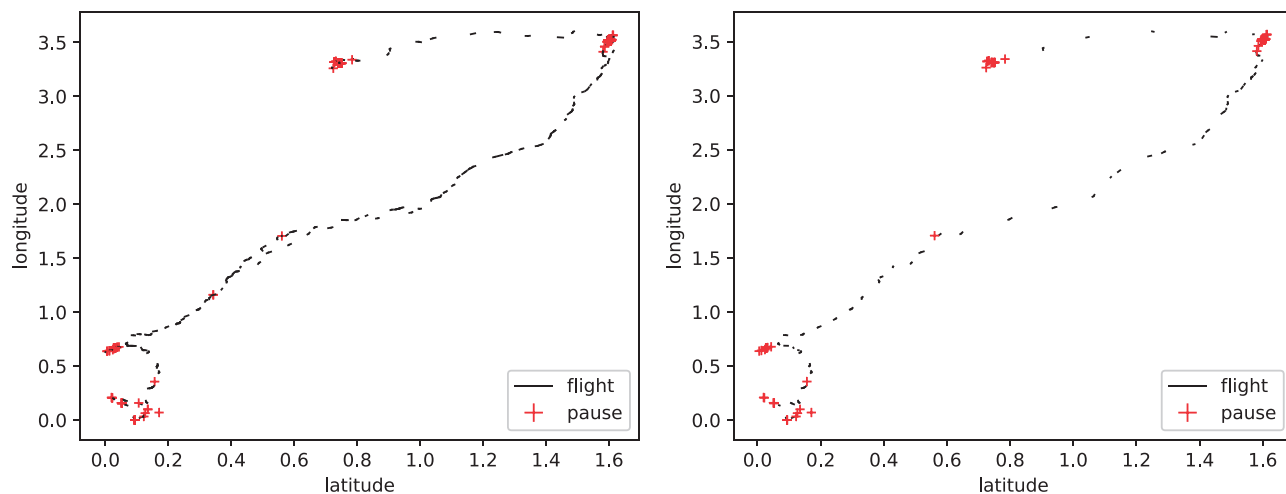


Figure 4. All 4185 observed flights and pauses of a subject during 5 months summarized by the rectangular method (left) and 320 selected flights and pauses from sparse online Gaussian Process (right). The original latitude and longitude are shifted by subtracting the corresponding minimal observed value to protect the user's privacy.

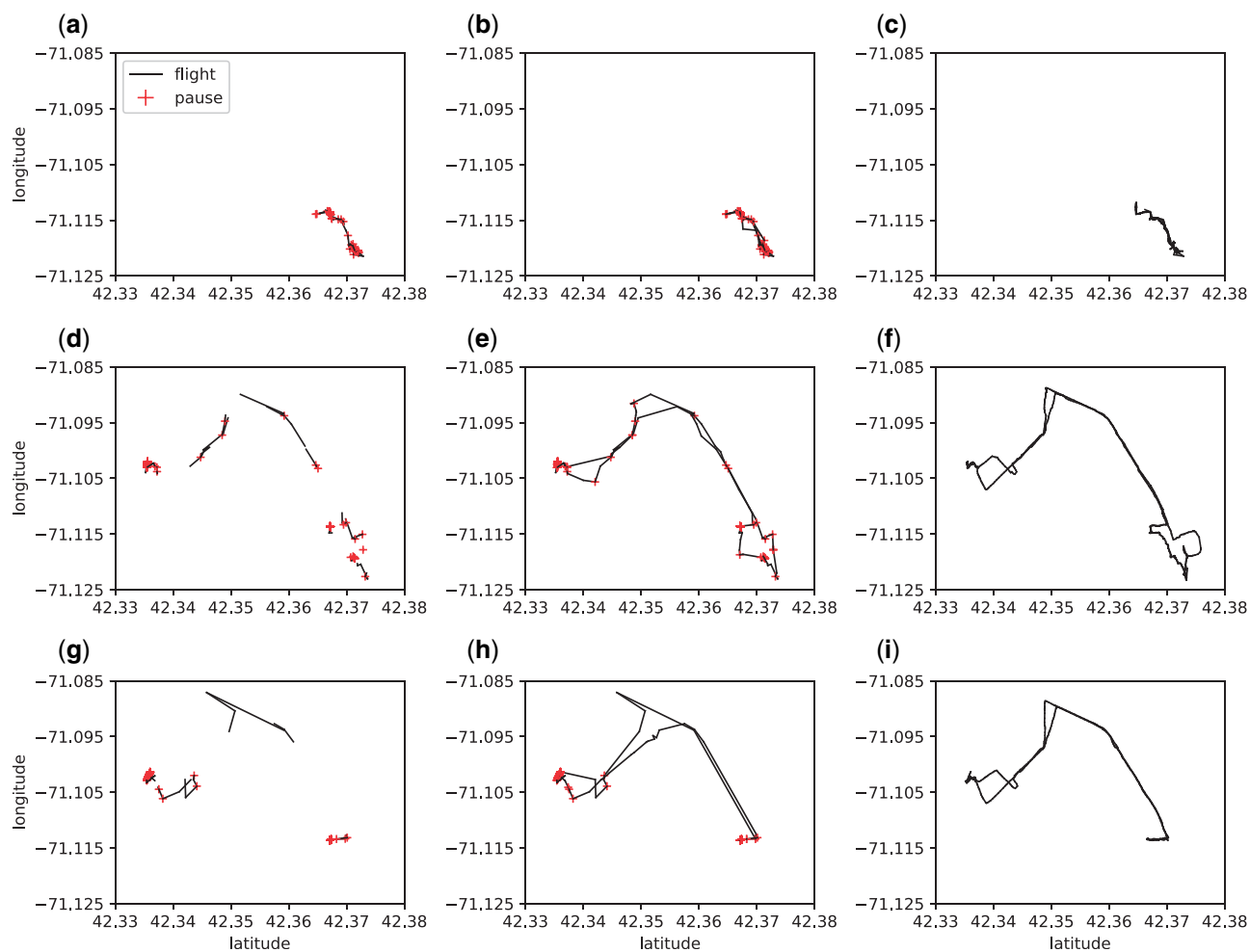


Figure 5. Comparisons among the artificial observed (a)(d)(g), imputed (b)(e)(h), and fully recorded trajectories (c)(f)(i) from real-world data collected over 3 days, in which each row shows the trajectories in a day. The fully recorded data were collected using a dedicated GPS tracker by G.L. The artificial observed data are created by randomly removing part of the fully observed data. The imputed trajectories are generated by the proposed method.

Table 1. Means and 95% CIs of the summary statistics derived from the imputed trajectories using a 10-time multiple imputation, compared with the truth derived from the fully collected data by the professional device

Time	Metric	Real-World Data	
		Truth	Estimate
Day 1	Home time	19.32	19.32 (19.25-19.39)
	Distance traveled	5.96	4.74 (4.73-4.75)
	Flight time	0.94	0.92 (0.71-1.13)
Day 2	Home time	13.16	13.09 (13.00-13.18)
	Distance traveled	16.06	14.06 (14.04-14.08)
	Flight time	1.44	1.63 (1.32-1.94)
Day 3	Home time	17.12	17.04 (16.96-17.12)
	Distance traveled	12.84	12.97 (12.95-12.99)
	Flight time	0.77	1.09 (0.82-1.36)

CI: confidence interval.

ment error. Next, we randomly select a starting time from 12 AM to 12:15 AM on each day, remove first $p \times 100\%$ of data every 15 minutes and treat the remaining as the observed data. The proposed imputation method is applied to the observed data and 2 metrics are derived from the imputed trajectories: the proportion of number of detected colocations events and the proportion of detected colocation duration. We replicate the procedure 50 times for every $p \in [0.1, 0.2, \dots, 0.9]$ and $h \in [15, 20, \dots, 35]$ and calculate the average of each metric both for regularly visited places and occasionally visited places. Figure 6 shows that imputation can significantly improve the power of detecting colocations, and the increase in power is greater for regularly visited locations than for occasionally visited locations. Based on this simulation, if 2 individuals are collocated

for 30 minutes and we would like to detect this event with a probability greater than 0.9 using the CDC guidelines, the GPS needs to be sampled for 12 minutes in every 15 minutes without imputation whereas it only needs to be sampled for 3 minutes every 15 minutes with imputation. The former sampling scheme would be guaranteed to deplete the phone's battery quickly, whereas the latter sampling scheme is practicable for detecting meaningful colocation in the context of COVID-19.

DISCUSSION

Despite the improvements in our new method, it imposes strong assumptions in some scenarios; for example, we interpolate the starting point and ending point when a long-distance airplane flight is detected. The direction adjustment function is subjective, which bends the flight toward the destination with high probability and almost rules out the possibility that the trajectory could start in the opposite direction. The modification for avoiding too frequent flight-pause switches may be unnecessary and cause bias in some scenarios. In general, our method simplifies the imputed trajectory by discarding probable small displacements, but it does capture the overall trajectory fairly precisely. One technical problem that could occur but is not addressed in our approach is the instability of GPS data during the first several seconds in a flight. The GPS receiver requires a short amount of time to locate the smartphone accurately, so the recorded latitude and longitude coordinates in the first few seconds are not reliable. Our approach addresses this issue with a smoothing function. Other improvements could be achieved by dividing a subject's activity area to multicenter areas if long flights are detected.

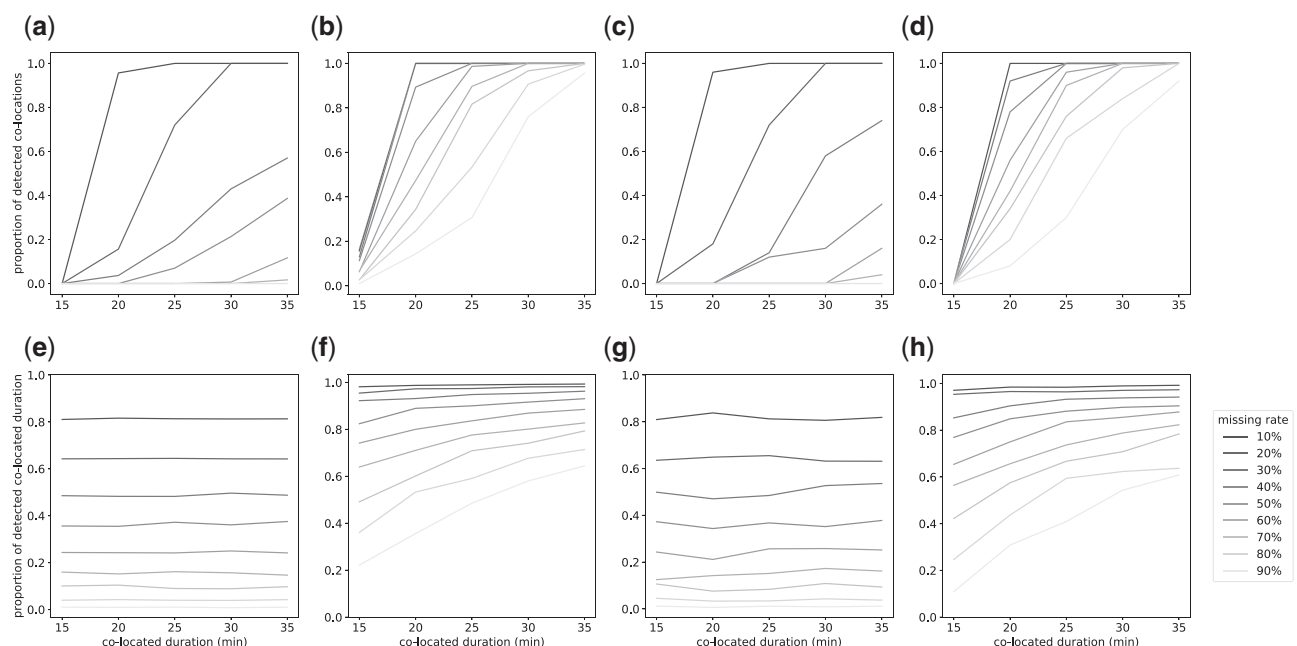


Figure 6. Comparison of the power to detect colocations given different rates of missing data. In this simulation, 2 individuals are collocated for h minutes (x-axis) either at a regularly visited place (a)(b)(e)(f) or at an occasionally visited place (c)(d)(g)(h) when $p \times 100\%$ of data are missing. Two metrics are calculated using the observed data (a)(c)(e)(g) and the imputed trajectories (b)(d)(f)(h). The first metric, displayed in the top row, is the proportion of detected colocation events out of the total number of colocations as defined by the Centers for Disease Control and Prevention guidelines; the second metric, displayed in the bottom row, is the proportion of detected colocation duration. For each value of h and p , a total of 50 different trajectories are generated independently and the missing data are imputed for each; the values on the y-axis are the corresponding means.

CONCLUSION

In this article, we describe a new method that augments our previous imputation method to overcome 2 of its important limitations: computational cost and accuracy. Our new method takes advantage of the continuous nature of human activity trajectories, uses a more natural spherical geometry rather than planar geometry to operationalize movement, adopts sparse online Gaussian Process to make the algorithm fast and scalable, and introduces some additional improvements, such as bidirectional imputation. We have demonstrated that our new method is substantially faster and has greater accuracy than the previous method.

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AUTHOR CONTRIBUTIONS

All authors participated in the design of the methods. GL implemented the methods in Python and conducted the experiments. GL and J-PO wrote the manuscript. J-PO supervised the research. All authors reviewed and revised the manuscript.

SUPPLEMENTARY MATERIAL

[Supplementary material](#) is available at *Journal of the American Medical Informatics Association* online.

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CONFLICT OF INTEREST STATEMENT

None declared.

DATA AVAILABILITY STATEMENT

We have implemented the method as a Python (version 3.6.6) package, which is available at <https://github.com/onnella-lab/forest/tree/master/forest/jasmine>.

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